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NFC – NDEF Formatting

An [NFC chip](#) must be first formatted as [NDEF](#) to be able to store NDEF messages and records in its user memory. NDEF formatting is a permanent operation.

Many applications of NFC do not use NDEF; these tend to be closed-loop systems such as secure access, loyalty, payment and transportation. These deployments should not use pre-NDEF formatted NFC chips.

Process

When a Type 2 NFC chip is NDEF formatted, special data is written to the NFC chip's Capability Container. These bytes start with *0xE1 0x10*. The Capability Container memory is [OTP](#), meaning that it can only be written to once. This is implemented via a bitwise *OR* operation. Therefore NDEF formatting is a permanent operation; once an NFC chip has been NDEF formatted it can not be unformatted.

Most NFC tag encoding systems, including the [GoToTags Encoder](#) will detect if an NFC tag needs to be NDEF formatted before trying to write to the tag, and if so, format it before writing the NDEF data.

NTAG Series

The NXP NTAG series of NFC chips (NTAG213, NTAG210 Micro, NTAG216...) come from the factory pre-NDEF formatted. The Capability Container of these chips has already been written to. NTAG chips can only be used for projects which require NDEF formatting.

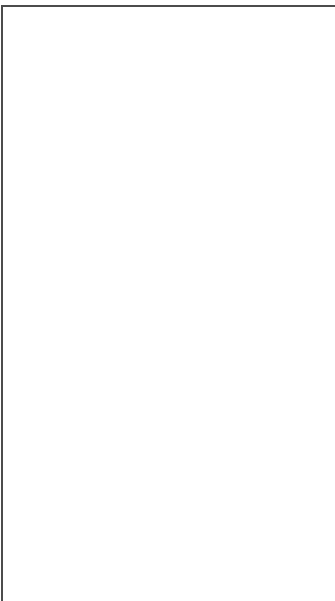
Memory Organization of NTAG213



NTAG213 Capability Container



Empty and NDEF Formatted NTAG



Systems

Many systems including [iOS](#) can only handle NDEF formatted NFC chips.

Updated on October 25, 2019

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